

UWASE SONIA

Full Stack Developer | AI & Robotics | UI/UX Design

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Professional Summary

Tech Enthusiast at Rwanda Coding Academy who enjoys turning real-world problems into working technology. Experienced in software development, Artificial Intelligence, robotics, embedded systems, computer vision, and UI/UX design. As a co-founder, robotics competitor, project builder, and mentor, I have learned to work across both technical and creative challenges. I am driven by curiosity, enjoy debugging and understanding complex systems, and look for opportunities to build practical, intelligent, and useful technology.

EDUCATION

Rwanda Coding Academy Rwanda

Sep 2024 - July 2027 Software Programming and Embedded Systems

College Du Christ Roi Rwanda

Sep 2021 - July 2024 O Level Studies

Saint Charles Lwanga Rwanda

Jan 2015 - July 2021 Primary Studies

TECHNICAL SKILLS

Programming

Java • Python • JavaScript • TypeScript • C • C++ •
PHP • Rust Basics

Software Development	React • Next.js • NestJS • Node.js • Spring Boot • Django • FastAPI
Robotics & Embedded	ESP32 • ESP8266 • ESP32-CAM • Microcontrollers • Sensors • Actuators • Relay Modules • IoT Systems
Databases & Backend	PostgreSQL • MySQL • MongoDB • Supabase • Firebase • REST APIs
DevOps & Tools Design	Git • GitHub • Docker • Linux • AWS Figma • UI/UX Design • Wireframing • Prototyping • User Flows • Design Systems

SOFT SKILLS

- **Communication & Feedback:** Communicate ideas clearly, actively listen to others, welcome constructive feedback, and use feedback to improve my work and approach.
- **Teamwork & Collaboration:** Work effectively with others, contribute ideas, respect different perspectives, and support teammates in achieving shared goals.
- **Problem Solving:** Approach challenges by understanding the root problem, exploring possible solutions, and finding practical ways forward.
- **Adaptability & Willingness to Learn:** Adapt to new environments, technologies, and responsibilities while continuously learning from experience and challenges.
- **Leadership & Initiative:** Take responsibility for tasks, contribute ideas, step forward when needed, and help turn ideas into action.

- **Critical Thinking:** Analyze problems carefully, question assumptions, and consider different approaches before making decisions.
- **Creativity & Innovation:** Generate ideas and explore new ways of using technology to address real-world problems.
- **Mentoring & Knowledge Sharing:** Share knowledge with others and help students develop their ideas, solve problems, and build prototypes.
- **Persistence:** Remain committed when facing difficult technical or project challenges and continue investigating until I understand the problem.
- **Organization & Time Management:** Manage multiple academic, technical, and extracurricular responsibilities while meeting deadlines and commitments.

PROFESSIONAL EXPERIENCE

RoboSpati — Co-Founder 2026-Present

AI & Robotics | robospati.com Rwanda

- Co-founded a technology initiative developing practical solutions with Artificial Intelligence and Robotics.
- Prototype robotics systems that combine microcontrollers, camera modules, sensors, servos, and software.
- Explore computer-vision applications such as person detection, facial landmarks, and robotic response.
- Work across the full development process, from problem identification and system design to hardware prototyping and implementation.

Impano Entertainment — Co-Founder 2026-Present

Creative production company | impanoentertainment.com Rwanda

- Co-founded a company working across cinematography, production design, creative direction, post-production, visual effects, drone visuals, color grading, sound design, commercials, documentaries, and narrative film.
- Contribute to the company's creative and technical direction and translate creative ideas into practical production workflows.

Freelance UI/UX Designer 2026

Freelance Rwanda

- Designed interfaces and user experiences for client projects, including a hotel website.
- Translated requirements into user flows, layouts, and prototypes, combining UI/UX knowledge with software-development experience.

Government / Public-Sector Technology Projects – Technology Contributor 2026

Professional development environment, Rwanda

- Contributed to technology projects addressing real-world public-sector requirements, working in project teams under professional constraints.
- Some project names, technical details, and organizational information are omitted due to confidentiality.

TECHNICAL PROJECTS

Impano Initiative – Digital Platform 2026

Technical Contributor / Developer, Rwanda

- Built a platform improving community access to information and enabling contributions toward school-feeding initiatives.

- Implemented user-facing interfaces and backend functionality using React/Next.js, PostgreSQL/Supabase, cloud storage, and payment integration.
- Integrated Cloudinary media delivery and multilingual support (English, Kinyarwanda, French); prepared the platform for public deployment.

SparkLock – Intelligent Hazard Detection System, 2nd Place, School Hackathon

Developer / Embedded Systems Contributor, Rwanda Coding Academy

- Contributed to a safety system that detects potential electrical and gas-related hazards before they lead to fires.
- Started on the mobile application, then expanded into embedded work with temperature sensors, relay modules, and buzzers.
- Explored how software and physical sensing/control components work together as one safety system.

Computer Vision & Robotic Tracking

Independent Research & Prototype

- Prototyped person detection and tracking with camera systems and ESP8266/ESP32-class hardware.
- Explored facial detection, landmark detection, and alignment so robotic systems can respond to a person's position or movement.
- Resolved camera input, Python environment, library, and hardware-communication issues through iterative debugging.

Choir Music Technology Platform

Independent Product Concept & Development | cantora.robospati.com

- Identified a recurring problem: choir rehearsals are difficult without a pianist or instrumental accompanist.
- Designed a platform concept for digital sheet music, composition, playback, and multiple instrument options, built around real choir workflows.

COMPETITIONS & ACHIEVEMENTS

FIRST LEGO League — Researcher & Secretary 2024

National Robot Game winner | SparkLock: 2nd Place, School Hackathon

- Team won the Robot Game at the national level, received a LEGO trophy, and earned the chance to compete internationally in Houston, Texas.
- Contributed to research, documentation, teamwork, and competition preparation.

Coderina STEM Summer Camp — Project Competition Winner 2024

Collaborative STEM project

- Developed a practical STEM project with a team and won the camp's project competition.

Ideation Coach — Year One Robotics Students | 2026

Student mentorship & solution development

- Coached Year One students through brainstorming, problem identification, solution development, and user-focused presentation, helping them turn ideas into practical solutions and communicate them effectively to their intended users.

LEADERSHIP & COMMUNITY

Embedded Systems Sessions — Rwanda Coding Academy

Session Lead

- Plan and run hands-on sessions on microcontrollers (Arduino Uno, ESP8266, ESP32), motors, and sensors, with Tinkercad demonstrations for prototyping before building in Robotics Club.

Rwanda Coding Academy Catholic Committee 2026-Present

Advisor

- Advise within the school's Catholic community and help organize community activities.

ADDITIONAL INFORMATION

Languages English • Kinyarwanda • French

Interests Cybersecurity

- Technology Entrepreneurship
- Product Design
- Music Technology
- STEM Education